98. $1 + 2 + 3 + \ldots + n = \dfrac{n(n+1)}{2}, \forall n \in \mathbb{N}^*.$

99. $2 + 5 + 8 + \ldots + \ldots + (2 + 3n) = \dfrac{(n+1)(4+3n)}{2}, \forall n \in \mathbb{N}.$

100. $2^0 + 2^1 + 2^2 + \ldots + 2^{n-1} = 2^n - 1, \forall n \in \mathbb{N}^*.$

101. $1^2 + 2^2 + 3^2 + \ldots + n^2 = \dfrac{n(n+1)(2n+1)}{6}, \forall n \in \mathbb{N}^*.$

102. $1^3 + 2^3 + 3^3 + \ldots + n^3 = \left[\dfrac{n(n+1)}{2}\right]^2, \forall n \in \mathbb{N}^*.$

103. $8 \mid (3^{2n} - 1), \forall n \in \mathbb{N}^*.$

99) BÁSICA:

$n = 0$

$2 + 3 \cdot 0 = \dfrac{(0+1)(4+3\cdot 0)}{2}$

$2 = 2$ OK

Hipótese indutiva:

Supor que $2 + 5 + _ + 2 + 3n = \dfrac{(n+1)(4+3n)}{2}$

Passo indutivo:

Provar que $2 + 5 _ + 2 + 3n + (2 + 3(n+1)) = \dfrac{(n+1+1)(4+3(n+1))}{2}$

$$\underbrace{2+5+_+2+3n}_{\text{Por hipótese}} + 5 + 3n = \frac{(n+2)(7+3n)}{2}$$

$$\frac{42}{9} = \frac{14}{3}$$

$$\frac{(n+1)(4+3n)}{2} + 5 + 3n = \frac{(n+1)(4+3n) + 2(5+3n)}{2}$$

$$= \frac{1}{2}\left((n+1)(3n+4) + 6n + 10\right) = \frac{1}{2}\left[3n^2 + 4n + 3n + 4 + 6n + 10\right]$$

$$= \frac{1}{2}\left[3n^2 + 13n + 14\right]\frac{3}{3} = \frac{3}{2}\left(\frac{3n^2}{3} + \frac{13n}{3} + \frac{14}{3}\right) = \frac{3}{2}\left(n^2 + \underset{S}{\frac{13n}{3}} + \underset{P}{\frac{14}{3}}\right)$$

multiplicando e dividindo por 3

$$= \frac{3}{2}\left[\left(n + \frac{7}{3}\right)\left(n + \frac{6}{3}\right)\right] = \frac{3}{2}\left[\left(n + \frac{7}{3}\right)(n+2)\right]$$

$$= \frac{1}{2}\left(3n + \frac{3 \cdot 7}{3}\right)(n+2) = \frac{1}{2}(n+2)(3n+7)$$

98. $1 + 2 + 3 + \ldots + n = \dfrac{n(n+1)}{2}, \forall n \in \mathbb{N}^*.$

99. $2 + 5 + 8 + \ldots + \ldots + (2 + 3n) = \dfrac{(n+1)(4+3n)}{2}, \forall n \in \mathbb{N}.$

100. $2^0 + 2^1 + 2^2 + \ldots + 2^{n-1} = 2^n - 1, \forall n \in \mathbb{N}^*.$

101. $1^2 + 2^2 + 3^2 + \ldots + n^2 = \dfrac{n(n+1)(2n+1)}{6}, \forall n \in \mathbb{N}^*.$

102. $1^3 + 2^3 + 3^3 + \ldots + n^3 = \left[\dfrac{n(n+1)}{2}\right]^2, \forall n \in \mathbb{N}^*.$

103. $8 \mid (3^{2n} - 1), \forall n \in \mathbb{N}^*.$

100) BASICA: $n=1$

$2^{(1-1)} = 2^1 - 1 \quad 1 = 1$ OK

Hipótese Indutiva:

Válido p/ $2^0 + _ + 2^{n-1} = 2^n - 1$

Passo indutivo: Para $n+1$

Provar que $\underbrace{2^0 + _ + 2^{n-1}}_{} + 2^{n+\cancel{1}-\cancel{1}} = 2^{n+1} - 1$

$2^n - 1 + 2^n =$

$= 2.2^n - 1 = 2^{n+1} - 1$, OK

101) BÁSICA: $n=1 \quad 1^2 = \dfrac{1(1+1)(2.1+1)}{6} \quad \therefore \quad 1 = \dfrac{1.\cancel{2}.\cancel{3}}{\cancel{6}} = 1$ OK

Hipótese indutiva: Supor válido p/ n

$1^2 + 2^2 + _ + n^2 = \dfrac{n(n+1)(2n+1)}{6}$

101. $1^2 + 2^2 + 3^2 + \ldots + n^2 = \dfrac{n(n+1)(2n+1)}{6}, \forall\, n \in \mathbb{N}^*.$

102. $1^3 + 2^3 + 3^3 + \ldots + n^3 = \left[\dfrac{n(n+1)}{2}\right]^2, \forall\, n \in \mathbb{N}^*.$

103. $8 \mid (3^{2n} - 1), \forall\, n \in \mathbb{N}^*.$

101) BÁSICA: $n = 1$ $\quad 1^2 = \dfrac{1(1+1)(2\cdot 1+1)}{6} \quad \therefore \; 1 = \dfrac{1\cdot \cancel{2}\cdot \cancel{3}}{\cancel{6}} = 1$ OK

Hipótese indutiva: Supor válido p/ n

$$1^2 + 2^2 + _ + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Passo indutivo p/ $n+1$: Provar que $\boxed{\overbrace{1^2+2^2+_+n^2}^{e} + (n+1)^2 = \dfrac{(n+1)(n+2)(2n+3)}{6}}$

$$\frac{3}{2} + \frac{4}{\cancel{2}} = \frac{7}{2}$$

$$\frac{3}{2} + 2 = \frac{7}{2}$$

$$\frac{n(n+1)(2n+1)}{6/1} + (n+1)^2 = \frac{n(n+1)(2n+1) + 6(n+1)^2}{6}$$

$$\frac{(n+1)}{6}\left[n(2n+1) + 6(n+1)\right] = \frac{(n+1)}{6}\left[2n^2 + n + 6n + 6\right]$$

$$\frac{(n+1)}{6}\left[2n^2 + 7n + 6\right] \times \frac{2}{2} = \frac{2(n+1)}{6}\left[n^2 + \frac{7}{2}n + \frac{\overset{3}{\cancel{6}}}{\cancel{2}}\right]$$

$$\frac{2(n+1)}{6}\left[\left(n + \frac{3}{2}\right)\left(n + \frac{\overset{2}{\cancel{4}}}{\cancel{2}}\right)\right] = \frac{2(n+1)}{6}\left(n + \overset{S}{\frac{3}{2}}\right)\left(n + \overset{P}{2}\right)$$

$$= \frac{(n+1)}{6}(2n+3)(n+2) = \frac{(n+1)(n+2)(2n+3)}{6}$$